Compactness first: a roadmap for formalizing the compactness theorems of geometric analysis and symplectic topology

A planning document, written for a research group or a lab with the compute to run large formalization efforts and looking for a target that matters. It argues that the compactness theorems of geometric analysis, and the frameworks built to tame them in symplectic and contact topology, are that target; takes inventory of what they are and what they need; and proposes an order of attack along two tracks. Statements about the current contents of the Lean library Mathlib are as of the author's knowledge in September 2026 and should be re-checked against the library before work begins.

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1. Why compactness

Nearly every existence theorem in geometric analysis has the same shape. One wants a solution of a nonlinear partial differential equation: a harmonic map, a minimal surface, a Yang–Mills connection, a metric of constant scalar curvature, a periodic orbit of a Hamiltonian system, a pseudoholomorphic curve, a solution of a nonlinear Schrödinger or wave equation. One writes down an energy whose critical points are the solutions, takes a sequence of approximate critical points (a minimizing sequence, a Palais–Smale sequence, a sequence of solutions with bounded energy), and asks whether it converges. In the natural function spaces it does *not* converge in general, and the theorems that say exactly *how* it fails are the compactness theorems:

- The sequence may **concentrate**: its energy collects at points, and after rescaling one sees a nontrivial solution on the whole space, a *bubble*. Sacks–Uhlenbeck for harmonic maps (1981), Brezis–Coron's "how to blow bubbles" (1985), Struwe's global compactness for the Yamabe-type functional (1984), Uhlenbeck's compactness for Yang–Mills (1982), the bubble trees of Gromov compactness (1985).
- The sequence may **vanish** or **split**: spread out, or separate into pieces that drift apart. Lions's concentration-compactness principle (1984–85) classifies this; the profile decompositions of Gérard (1998) and Bahouri–Gérard (1999) make it quantitative.
- In variational problems the failure of compactness itself has a **Morse theory**: the lost sequences behave like critical points that are not there, and their topology is computable. This is Bahri's theory of *critical points at infinity* (1989), which with Coron (1988) proved that a domain with nontrivial topology carries a solution of the critical-exponent equation that has no positive solution on a ball, and which Bahri and Lions (1992) tied to Morse indices of Palais–Smale sequences. The same viewpoint drives Bahri's variational approach to periodic orbits of contact forms and to the Weinstein conjecture.

- In symplectic and contact geometry, **Hofer's compactness** for finite-energy pseudoholomorphic planes in a symplectization (1993) turned the Weinstein conjecture in dimension three into a bubbling-off analysis; Hofer, Wysocki and Zehnder then described the asymptotics of such curves near punctures (1995–2000), and Bourgeois, Eliashberg, Hofer, Wysocki and Zehnder proved the **SFT compactness theorem** (2003): sequences of finite-energy curves converge to *holomorphic buildings*, stacked levels joined along Reeb orbits. This is the compactness theorem behind contact homology and symplectic field theory.
- The compact moduli spaces produced by Gromov and SFT compactness are not manifolds, and the invariants built from them need a *virtual fundamental cycle*. The frameworks for this, Fukaya– Oh–Ohta–Ono's Kuranishi structures, McDuff–Wehrheim's Kuranishi atlases, Pardon's implicit atlases, and Hofer–Wysocki–Zehnder's **polyfold theory**, are thousands of pages of analysis whose purpose is precisely to put a smooth, Fredholm structure on a compactified moduli space so that compactness and transversality can be used together. The correctness of these foundations was publicly debated in the 2010s; it was settled, but by argument and re-exposition, not by anything a machine has checked.
- In the geometric setting the *domain itself* may degenerate: Cheeger–Gromov compactness for Riemannian manifolds, Hamilton's version for Ricci flows (1995), Perelman's no-local-collapsing; and in geometric measure theory Federer–Fleming's compactness for integral currents (1960) and Allard's regularity (1972) are what make minimal surfaces exist.

These theorems are what make the subjects work: knowing the *only* ways convergence can fail, one rules them out by an energy estimate, a topological argument, or a virtual count, and the solution or the invariant exists. Yamabe (Aubin, Schoen), Bahri–Coron, Sacks–Uhlenbeck, Donaldson theory, Floer homology, Gromov–Witten theory, contact homology, the Weinstein conjecture (Hofer in dimension three for overtwisted and S^3-like cases, Taubes in general), Perelman's geometrization, the Kenig–Merle theory of critical dispersive equations, all run through a compactness theorem.

At the time of writing, none of these theorems is formalized, in Lean or in any other proof assistant, and the infrastructure they rest on (Sobolev spaces, elliptic regularity, analysis on Riemannian manifolds, the sc-calculus) is largely absent from Mathlib. The headline results recently announced with machine-checked proofs sit in other parts of analysis. This document is a case that compactness is the right next place to build, and a plan for doing it.

Three reasons this is the right target rather than a single famous theorem. First, *leverage*: one compactness theorem is a dependency of dozens of existence theorems. Second, *honesty of statement*: compactness theorems have clean, checkable statements which a human mathematician can verify without reading the proof; that is what a large automated effort needs from its human collaborators,

and it is the part that does not need compute. Third, *unfinished verification*: the symplectic foundations in particular are long, were contested, and are relied upon by a whole field; a machine-checked polyfold theory would be a genuine service, not a demonstration.

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2. The tally

Each entry gives: the statement in one or two sentences, the source, what it needs, what needs it, the state of the Lean library Mathlib as far as the author knows (marked *check*), notes for the formalizer, and a size: **S** comparable to Sard's theorem in the accompanying Milnor project (one working session with an agent, about 1,500 lines including missing infrastructure), **M** a few such units, **L** months for a team with agents, **XL** a multi-year library effort. Entries are grouped in blocks A–G; the dependency graph in §3 uses the labels.

A. Functional-analytic core

T1. Arzelà–Ascoli. A subset of $C(K, Y)$, K compact, is precompact iff it is pointwise precompact and equicontinuous. *Mathlib:* present (`BoundedContinuousFunction.arzela_ascoli`). *Needed by:* every compactness argument that passes to C^k convergence on compact sets. **Done.**

T2. Banach–Alaoglu and weak sequential compactness. The closed unit ball of the dual of a normed space is weak-* compact; in a reflexive space, bounded sequences have weakly convergent subsequences; in a Hilbert space this is elementary. *Mathlib:* Banach–Alaoglu present (`WeakDual.isCompact_polar`); the sequential statement for Hilbert spaces should be checked and, if missing, is S. *Needed by:* T8, S6, every minimizing-sequence argument. **Done / S.**

T2′. Gromov–Hausdorff precompactness. A family of compact metric spaces with a uniform bound on the number of ε-balls needed to cover, for each ε, is precompact in the Gromov–Hausdorff metric. *Source:* Gromov (1981). *Mathlib:* the Gromov–Hausdorff space and its completeness are present (`GromovHausdorff`), including a total-boundedness criterion (*check* the exact form). *Needed by:* T14, Cheeger–Colding theory. *Note:* this is the one deep compactness theorem of the subject that is essentially already formalized; it should be advertised as such. **Done.**

D1. Distributions, weak derivatives, mollifiers, cutoffs. Test functions, the distributional derivative, convolution with an approximate identity converging in L^p, smooth bump functions with prescribed support. *Mathlib:* Schwartz space, smooth bump functions and convolution are present; general distributions on open sets are partial (*check*). *Needed by:* T3 and everything after. *Note:* decide once whether $W^{k,p}$ is built on distributions or on the closure of smooth functions; the two are equivalent (Meyers–Serrin) but the proof of equivalence is itself an M-sized task. **S–M.**

T3. Sobolev spaces $W^{k,p}(\Omega)$. Definition via weak derivatives; completeness; density of C_c^∞ in $W^{k,p}(\mathbb{R}^n)$ and of $C^\infty(\bar{\Omega})$ for Lipschitz Ω; extension operator for Lipschitz domains; trace theorem; the homogeneous space $\dot{H}^1$. *Source:* any of Adams–Fournier, Evans, Brezis. *Mathlib:* absent as a theory (*check* for recent work). *Needed by:* T4–T14, V1–V4, P1–P3, F4. *Notes:* work over L^p equivalence classes; state for open subsets of a finite-dimensional inner product space, not $\mathbb{R}^n$; the extension and trace theorems are the expensive parts and are not needed for the $\mathbb{R}^n$ theory, so they can be deferred to after T8. **M (for $\mathbb{R}^n$), L (with domains).**

T4. Sobolev, Gagliardo–Nirenberg, Morrey, Hardy–Littlewood–Sobolev inequalities. $\|u\|_{L^{p^*}} \le C\|\nabla u\|_{L^p}$ for $p < n$, $p^* = np/(n-p)$; Hölder continuity for $p > n$; sharp constants (Aubin, Talenti 1976; Lieb 1983 for HLS). *Needed by:* T5, T9, T10, V2. *Notes:* the Gagliardo trick (integrate along lines, then Hölder) formalizes cleanly; the sharp constant and the classification of extremals (the bubbles $U(x) = (1+|x|^2)^{-(n-2)/2}$ up to scaling and translation) are a separate M task needed by T10 and V2. **M.**

T5. Rellich–Kondrachov. For bounded Lipschitz Ω and $q < p^*$, the inclusion $W^{1,p}(\Omega) \to L^q(\Omega)$ is compact. *Needs:* T3, T4, and the Kolmogorov–Riesz–Fréchet criterion: a bounded subset of $L^p(\mathbb{R}^n)$ is precompact iff it is tight and $\sup_f \|f(\cdot + h) - f\|_p \to 0$ as $h \to 0$. *Needed by:* T8–T14, V1–V3. *Note:* prove Kolmogorov–Riesz first as a standalone theorem; it is reusable and S. **M.**

E2. Fredholm theory and the implicit function theorem in Banach spaces. Fredholm operators, index, stability under small and compact perturbations, the C^k implicit function theorem, the Sard–Smale theorem (regular values of Fredholm maps of index d are generic). *Mathlib:* the Banach IFT is present; Fredholm operators and Sard–Smale are absent (*check*). *Needed by:* V1, F3, S6, and every transversality argument. *Note:* Sard–Smale needs Sard's theorem, which the Milnor project has. **S–M.**

E1. Elliptic regularity. Interior and boundary L^p estimates for the Laplacian (Calderón–Zygmund), Schauder estimates, ε-regularity for the equations in T10–T12, S1 ($\|\nabla u\|_{L^2}$ small on a disc implies C^k bounds inside). *Needed by:* T10–T14, SW, S1–S7. *Notes:* the single largest infrastructure item in Track I; nothing geometric can be completed without it; the Calderón–Zygmund theory needs singular integrals, which Mathlib does not have. Start with the L^2 theory (energy estimates, difference quotients) which suffices for ε-regularity of harmonic maps and J-curves in two dimensions, and defer L^p. **L.**

B. Variational methods and critical points at infinity

V1. Palais–Smale condition, deformation lemma, mountain pass and linking theorems. If J is C^1 on a Banach space, satisfies $(\mathrm{PS})_c$, and has a mountain-pass geometry, then c is a critical value. *Sources:* Palais–Smale (1964), Ambrosetti–Rabinowitz (1973), Rabinowitz's CBMS notes (1986); Ekeland's variational prin-

ciple (1974) as the alternative route. *Needs:* E2, pseudo-gradient vector fields and their flows (ODE in Banach spaces, in Mathlib). *Needed by:* T10, V2–V4, S2. *Note:* Ekeland's principle is a pure metric-space theorem and is S on its own. **M.**

T6. Lions's concentration-compactness lemma I. Let μ_k be probability measures on $\mathbb{R}^n$. Then, along a subsequence, exactly one of: *compactness* (there are y_k with $\mu_k(B(y_k,R)) \geq 1-\varepsilon$ for $R = R(\varepsilon)$), *vanishing* ($\sup_y \mu_k(B(y,R)) \to 0$ for every R), or *dichotomy* (there is $\alpha \in (0,1)$ and $\mu_k \approx \mu_k^1 + \mu_k^2$ with masses α, $1-\alpha$ and supports drifting apart). *Source:* Lions, Ann. IHP 1 (1984), Part 1, Lemma I.1. *Needs:* measure theory only. *Needed by:* T8, and conceptually everything in block B. *Note:* the proof is a diagonal argument on the concentration functions $Q_k(R) = \sup_y \mu_k(B(y,R))$; the formal statement must fix which subsequence and which cutoffs, which is exactly where textbook versions are loose. **S.**

T7. Brezis–Lieb lemma. If $u_k \to u$ a.e. and $\sup_k \|u_k\|_p < \infty$ then $\|u_k\|_p^p - \|u_k - u\|_p^p \to \|u\|_p^p$. *Source:* Brezis–Lieb, Proc. AMS 88 (1983). *Needs:* L^p, Fatou, dominated convergence. *Needed by:* T8, T9, T10. **S.**

T8. Lions's concentration-compactness II (locally compact case). For translation-invariant minimization problems $I_\lambda = \inf\{E(u) : \int |u|^q = \lambda\}$, every minimizing sequence is relatively compact up to translation iff the strict subadditivity $I_\lambda < I_\alpha + I_{\lambda-\alpha}$ holds for all $0 < \alpha < \lambda$. *Source:* Lions (1984), Parts 1 and 2. *Needs:* T5, T6, T7. *Needed by:* existence of ground states for nonlinear Schrödinger and Choquard equations, the Thomas–Fermi and Hartree problems. **M.**

T8′. The limit case: Lions's Rev. Mat. Iberoam. papers (1985). Concentration-compactness for critical-exponent problems, with the measure-theoretic lemma: if $u_k \rightharpoonup u$ in $\dot H^1$, $|\nabla u_k|^2 \rightharpoonup \mu$, $|u_k|^{2^*} \rightharpoonup \nu$, then $\nu = |u|^{2^*} + \sum_j \nu_j \delta_{x_j}$, $\mu \geq |\nabla u|^2 + \sum_j \mu_j \delta_{x_j}$, with $\nu_j^{2/2^*} \leq S^{-1}\mu_j$. *Needs:* T4 (sharp constant), T7. *Needed by:* T10, V2. **M.**

T9. Profile decomposition. Every bounded sequence in $\dot H^1(\mathbb{R}^n)$ has a subsequence $u_k = \sum_{j \leq J} \lambda_{j,k}^{-(n-2)/2}\, \phi_j((x - x_{j,k})/\lambda_{j,k}) + r_k^J$ with pairwise orthogonal scales/centres and $\limsup_k \|r_k^J\|_{L^{2^*}} \to 0$ as $J \to \infty$, together with the Pythagorean expansion of the $\dot H^1$ norm. *Sources:* Gérard, ESAIM COCV 3 (1998); Bahouri–Gérard, Amer. J. Math. 121 (1999) for the wave equation; Keraani for Strichartz norms. *Needs:* T3, T4, T7. *Needed by:* T10, P2, P3. *Note:* prove by induction on the energy defect; the orthogonality of scales is a clean lemma about sequences of pairs (λ, x). **M–L.**

T10. Struwe's global compactness; Brezis–Coron. Every Palais–Smale sequence for $E(u) = \frac{1}{2}\int |\nabla u|^2 - \frac{1}{2^*}\int |u|^{2^*}$ on a bounded domain is, up to a subsequence, a solution plus finitely many bubbles (rescaled solutions

on $\mathbb{R}^n$) with energies adding up. Brezis–Coron: the analogous statement for H-systems (constant mean curvature surfaces), the first "bubbling" theorem stated as such. *Sources:* Struwe, Math. Z. 187 (1984); Brezis–Coron, CPAM 37 (1984), CMP 92 (1985); Brezis–Nirenberg, CPAM 36 (1983) for the subcritical perturbation. *Needs:* T5, T8′, T9, E1, the Pohozaev identity, the classification of positive solutions on $\mathbb{R}^n$ (Gidas–Ni–Nirenberg, Caffarelli–Gidas–Spruck; itself an M–L task via the moving plane method). *Needed by:* V2, V3, and every "energy quantization" argument. **L.**

V2. Bahri–Coron: critical points at infinity. If a bounded domain $\Omega \subseteq \mathbb{R}^n$ has $H_d(\Omega; \mathbb{Z}/2) \neq 0$ for some $d \geq 1$, then $-\Delta u = u^{(n+2)/(n-2)}$, $u > 0$, $u|_{\partial\Omega} = 0$ has a solution. The proof studies the sublevel sets of E on the unit sphere of $\dot{H}^1_0$, shows via T10 that the only obstruction to compactness is the "critical points at infinity" (configurations of bubbles at points of Ω), computes their contribution to the topology of the level sets by an explicit expansion of E and its gradient near such configurations (the *Morse lemma at infinity*), and derives a contradiction from the topology of Ω. *Sources:* Bahri–Coron, CPAM 41 (1988); Coron, CRAS 299 (1984) for domains with a small hole; Bahri, *Critical points at infinity in some variational problems*, Pitman 182 (1989), for the general theory; Rey, JFA 89 (1990) for the role of the Green function. *Needs:* T10, V1, T4 with sharp constants and the explicit bubbles, singular homology of the domain and of level sets (Mathlib has singular homology only in nascent form; *check*), the expansion estimates. *Needed by:* V3, the Bahri–Coron scalar curvature theorem on S^3 (JFA 95, 1991), Bahri's contact form theory. *Notes:* this is the flagship of Track I because it is exactly the kind of proof a machine should check: long explicit expansions with error terms whose constants depend on the configuration, and a topological argument that must be stated precisely. Plan the algebraic-topology input separately. **L.**

V3. Bahri–Lions: Morse indices of Palais–Smale sequences. For superlinear subcritical problems $-\Delta u = f(u)$, bounds on the Morse index of a sequence of solutions give L^∞ bounds, hence compactness; conversely the Morse index of critical points obtained by min-max is controlled. *Sources:* Bahri–Lions, CPAM 45 (1992); Bahri–Lions, Ann. IHP 14 (1997) for unbounded domains. *Needs:* T10-type blow-up analysis, spectral theory of Schrödinger operators (Mathlib has the spectral theorem for compact self-adjoint operators; *check* for unbounded operators), Liouville theorems. **M–L.**

V4. Bahri's variational theory of contact forms; Bahri–Rabinowitz. The action functional on the space of Legendrian-type curves, its critical points (Reeb periodic orbits) and its critical points at infinity (pseudo-orbits); Bahri–Rabinowitz on periodic solutions of Hamiltonian systems of three-body type via critical points at infinity. *Sources:* Bahri, *Pseudo-orbits of contact forms* (1988); *Flow lines and algebraic invariants in contact form geometry* (2003); Bahri–Rabinowitz, Ann. IHP 8 (1991). *Needs:* V1, Hilbert manifolds of loops, ODE, contact geometry (absent). *Note:* this is where Track I meets Track II from the variational side; it is the road not taken by most of the field after Hofer 1993, and precisely for that

reason it is under-verified. **L.**

V5. Compactness of the Yamabe problem (Schoen's conjecture). The set of solutions of the Yamabe equation on a compact manifold not conformally equivalent to the sphere is compact in C^2, for $n \leq 24$; false for $n \geq 25$. *Sources:* Schoen (1991) conjecture and the locally conformally flat case; Li–Zhu (1999) $n = 3$; Druet (2004) $n \leq 5$; Marques (2005) $n \leq 7$; Li–Zhang (2005–07) $n \leq 11$; Khuri–Marques–Schoen, JDG 81 (2009) $n \leq 24$; Brendle, JAMS 21 (2008) counterexamples $n \geq 52$; Brendle–Marques, JDG 81 (2009) $25 \leq n \leq 51$. *Needs:* T10-type blow-up analysis on manifolds (G1, E1), the positive mass theorem (itself XL), Weyl tensor expansions. *Note:* the cleanest example of a compactness statement whose truth depends on the dimension in a nontrivial way; the counterexamples are explicit and would be a striking formalization target on their own (they are finite-dimensional calculus). **XL (theorem), L (counterexamples).**

C. Dispersive equations

P1. Strichartz estimates and critical local well-posedness. $\|e^{it\Delta} f\|_{L_t^q L_x^r} \leq C\|f\|_{L^2}$ for admissible (q, r); local existence for energy-critical NLS in $\dot{H}^1$. *Sources:* Strichartz (1977), Ginibre–Velo, Keel–Tao (1998). *Needs:* Fourier transform on L^2 (Mathlib has the Fourier transform on Schwartz space and L^2 Plancherel; *check*), the TT^* argument, dispersive estimate. **M.**

P2. Kenig–Merle: concentration-compactness and rigidity. For the focusing energy-critical NLS in dimensions $3, 4, 5$, radial data with energy below the ground state's scatter; proof: if not, a minimal-energy blow-up solution exists whose orbit is precompact modulo scaling (by T9 and P1), and a localized virial identity shows no such solution exists. *Sources:* Kenig–Merle, Invent. Math. 166 (2006); Acta Math. 201 (2008) for the wave equation. *Needs:* T9, P1, the virial identity. *Needed by:* the whole "induction on energy" and "minimal counterexample" method (Bourgain 1999; Colliander–Keel–Staffilani–Takaoka–Tao 2008; Killip–Visan). *Note:* the structure (extract a minimal almost-periodic counterexample, then exclude it) is the same as V2's and S2's; a formalization of the extraction step would serve all three. **L.**

P3. Merle–Raphaël: blow-up for the L^2-critical NLS. Solutions slightly above the ground-state mass with negative energy blow up with the log-log rate, and the blow-up profile is the ground state. *Sources:* Merle–Raphaël, Ann. Math. 161 (2005) and the series 2003–2006. *Needs:* P1, modulation theory, the spectral property of the linearized operator (checked numerically in the original; later proved rigorously in low dimensions). *Note:* included because it is the paradigm for the recent blow-up results; the numerically verified spectral input is a natural thing to formalize with interval arithmetic. **L–XL.**

D. Geometric analysis and gauge theory

G1. Analysis on Riemannian manifolds. Riemannian metrics, volume, Levi-Civita connection, Laplace–Beltrami, Sobolev spaces on compact manifolds, elliptic regularity there, harmonic coordinates (Jost–Karcher 1982; Anderson 1990). *Mathlib:* smooth manifolds, vector bundles and connections are present; Riemannian metrics are recent and partial; no Laplacian (*check*). *Needed by:* T11–T14, V5, S1–S7. **L.**

T11. Sacks–Uhlenbeck; bubble trees; energy identity. A sequence of harmonic maps (or α-harmonic approximations) from a closed surface with bounded energy has a subsequence converging in C^∞ away from finitely many points, where bubbles (harmonic spheres) form; the energy is conserved in the limit (Ding–Tian 1995; Parker 1996 bubble tree with necks). Existence of minimizing harmonic spheres in a manifold with nontrivial π_2. *Sources:* Sacks–Uhlenbeck, Ann. Math. 113 (1981); Parker, JDG 44 (1996); Ding–Tian, CAG 3 (1995); Schoen–Uhlenbeck, JDG 17 (1982) for regularity in higher dimensions; Hélein (1990) for regularity in two dimensions; Rivière, Invent. Math. 168 (2007) for conservation laws. *Needs:* E1 (ε-regularity, removable singularities), G1, T5. *Needed by:* S1 (same analysis with the Cauchy–Riemann operator replacing the harmonic map operator). **L.**

G2. Principal bundles, connections, curvature, gauge transformations, characteristic classes. *Mathlib:* vector bundles yes, principal bundles and gauge groups no (*check*). **M–L.**

T12. Uhlenbeck compactness; Coulomb gauge; removable singularities. Connections on a bundle over a compact n-manifold with $\|F_A\|_{L^p} \le C$, $p > n/2$, have a subsequence which, after gauge transformations, converges weakly in $W^{1,p}$; a connection with small $L^{n/2}$ curvature on a ball admits a Coulomb gauge with $W^{1,p}$ control; a finite-energy Yang–Mills connection on a punctured 4-ball extends. *Sources:* Uhlenbeck, CMP 83 (1982), two papers; Sedlacek (1982) for the weak compactness with bubbling; Wehrheim's book *Uhlenbeck Compactness* (2004) for a complete exposition, which is the text to formalize. *Needs:* E1, G2, T4. *Needed by:* Donaldson theory, Taubes' gluing, Floer's instanton homology. **L–XL.**

T12′. Tian's compactness for Yang–Mills in higher dimensions. Sequences of Yang–Mills connections with bounded energy on an n-manifold, $n \ge 5$, converge away from a closed set of codimension at least four which is rectifiable. *Source:* Tian, Ann. Math. 151 (2000). *Needs:* T12, GMT. **XL.**

SW. Seiberg–Witten compactness. For a closed 4-manifold with a spin^c structure, the moduli space of solutions of the Seiberg–Witten equations is compact; the proof is an a priori C^0 bound on the spinor from the maximum principle and the Weitzenböck formula, followed by elliptic bootstrapping. *Sources:* Witten (1994); Kronheimer–Mrowka (1994); Morgan's book (1996). *Needs:* G2, spinors and Dirac operators, E1 (light: L^2 theory and bootstrapping suffice). *Needed*

by: S7, Seiberg–Witten invariants. *Note:* the cheapest compactness theorem in gauge theory and the best first geometric target in block D after T11; no bubbling analysis at all. **M–L.**

GMT. Federer–Fleming compactness; Allard regularity. Integral k-currents in $\mathbb{R}^n$ with bounded mass and boundary mass form a compact set in the flat norm topology (Federer–Fleming 1960; the closure theorem). A stationary integral varifold whose density is close to one in a ball is a $C^{1,\alpha}$ graph there (Allard 1972). *Needs:* Hausdorff measures (in Mathlib), rectifiable sets, the deformation theorem, area and coarea formulas (absent). *Needed by:* existence of minimal surfaces (Plateau's problem), Almgren–Pitts min-max, De Lellis–Spadaro regularity theory. **L–XL.**

GMT′. Compactness for minimal surfaces. Choi–Schoen (1985): embedded minimal surfaces of fixed genus in a 3-manifold with positive Ricci curvature form a compact set in C^k. Colding–Minicozzi: compactness for embedded minimal disks without area bounds. *Needs:* GMT, G1, E1. **L.**

T14. Cheeger–Gromov and Hamilton compactness. Riemannian n-manifolds with $|\mathrm{Rm}| \leq K$, injectivity radius $\geq i_0$ and diameter $\leq D$ subconverge in $C^{1,\alpha}$ to a $C^{1,\alpha}$ manifold, up to diffeomorphism (Cheeger 1970; Gromov 1981; Greene–Wu, Peters 1987 for the $C^{1,\alpha}$ statement via harmonic coordinates, Anderson 1990); with bounds on all derivatives of curvature, smooth convergence, and Hamilton's version (1995) for sequences of Ricci flows. Cheeger–Colding (1997–2000): structure of limits under Ricci lower bounds alone. *Needs:* G1 (harmonic radius), E1, T2′. *Needed by:* Perelman's work, Ricci flow with surgery, the study of Ricci limit spaces. **XL.**

E. Symplectic and contact topology

S1. Gromov compactness. A sequence of J_k-holomorphic maps from closed Riemann surfaces of fixed genus to a compact symplectic manifold with $J_k \to J$ and bounded energy has a subsequence converging, in the Gromov topology, to a stable J-holomorphic map (a nodal curve with a map, with finite automorphism group). *Sources:* Gromov, Invent. Math. 82 (1985); Ye, Trans. AMS 342 (1994); Parker–Wolfson, JGA 3 (1993); Hummel, *Gromov's compactness theorem for pseudo-holomorphic curves* (1997); McDuff–Salamon, *J-holomorphic curves and symplectic topology*, Ch. 4–5 (2004). *Needs:* T11-type ε-regularity and removable singularities for the Cauchy–Riemann operator, the monotonicity lemma, the Deligne–Mumford compactification of the moduli of stable curves (algebraic geometry or hyperbolic geometry input, absent), G1. *Needed by:* S2, S4–S6, F4, F5, every Gromov–Witten or Floer-type invariant. *Notes:* the waist of Track II. Formalize McDuff–Salamon's exposition, which separates the analysis (Chapter 4) from the bookkeeping of nodal curves (Chapter 5). The genus-zero case needs no Deligne–Mumford theory beyond trees of spheres and should be done first. **XL (L for genus zero).**

S1′. Gromov's nonsqueezing theorem. A symplectic embedding $B^{2n}(r) \to$

$B^2(R) \times \mathbb{R}^{2n-2}$ exists only if $r \leq R$. *Proof:* S1 in genus zero on $S^2 \times T^{2n-2}$ plus the monotonicity lemma. *Note:* the natural first application of S1 with a headline statement. **L (given S1 genus zero).**

S2. Hofer: finite-energy planes and the Weinstein conjecture in dimension three. For a contact form on a closed 3-manifold with an overtwisted contact structure, or with $\pi_2 \neq 0$, or on S^3, the Reeb vector field has a periodic orbit. *Proof:* study J-holomorphic planes in the symplectization $\mathbb{R} \times M$ with finite Hofer energy; show by a bubbling-off analysis that a family of holomorphic discs with boundary on an overtwisted disc (Bishop family) must break, producing a finite-energy plane; show that a finite-energy plane is asymptotic to a periodic Reeb orbit. *Source:* Hofer, Invent. Math. 114 (1993). *Needs:* S1-type analysis on non-compact domains with the energy identity for the symplectization, V1-type geometry for the Bishop family, contact geometry, the asymptotic analysis of S3 (in its first form). *Needed by:* S3, S4, S7. *Note:* the compactness-plus-bubbling structure is identical to T10 and P2; the new ingredient is the non-compact target and the energy that controls it. **L–XL.**

S3. Hofer–Wysocki–Zehnder asymptotics; finite-energy foliations. A finite-energy J-holomorphic half-cylinder in a symplectization converges exponentially to a Reeb orbit, with leading term an eigenfunction of the asymptotic operator $-J_0\partial_t - S(t)$; consequences for the intersection theory of such curves; existence of finite-energy foliations of S^3 and of global surfaces of section for dynamically convex Reeb flows. *Sources:* HWZ, Ann. IHP 13 (1996); GAFA 5 (1995); Ann. Math. 148 (1998) for strictly convex energy surfaces; Ann. Math. 157 (2003) for finite energy foliations; Siefring (2008–11) for relative asymptotics. *Needs:* S2, spectral theory of the asymptotic operator (unbounded self-adjoint on $L^2(S^1)$; *check* Mathlib). *Needed by:* S4, S7, embedded contact homology. **L.**

S4. SFT compactness. Let (W, ω) be a symplectic cobordism with cylindrical ends and J cylindrical at infinity. A sequence of finite-energy punctured J-holomorphic curves of bounded energy and bounded genus has a subsequence converging to a *holomorphic building*: finitely many levels, each a nodal punctured curve in W or in a symplectization of an end, matched along Reeb orbits at the punctures, with the total energy and topology controlled. *Source:* Bourgeois– Eliashberg–Hofer–Wysocki–Zehnder, Geom. Topol. 7 (2003); Cieliebak–Mohnke, JSG 3 (2005) for the compactness in the Deligne–Mumford setting with the Reeb dynamics not Morse–Bott. *Needs:* S1 (with punctures), S2, S3, Deligne–Mumford with marked points and the hyperbolic-geometry description of degenerations (thick–thin decomposition), the "level structure" bookkeeping. *Needed by:* contact homology, SFT (Eliashberg–Givental–Hofer 2000), F4 for SFT. *Notes:* the summit of the compactness track; the statement itself is a page long and its precise formulation (the topology on the space of buildings) is one of the things a blueprint must settle. **XL.**

S5. Hofer–Zehnder capacity; energy–capacity inequality. Definition of c_{HZ} via Hamiltonians with no fast periodic orbits; $c_{HZ}(B^{2n}(r)) = \pi r^2$; the displacement energy bounds the capacity. *Sources:* Hofer–Zehnder (1990, book

1994); Hofer's displacement energy (1990); Lalonde– McDuff (1995) via S1. *Needs:* V1-type variational theory for the action functional (the original proof) or S1 (the holomorphic proof). **L.**

S6. Floer compactness and gluing. Solutions of the Floer equation $\partial_s u + J(\partial_t u - X_H(u)) = 0$ on the cylinder with bounded energy converge, modulo s-translation, to broken trajectories; gluing produces a trajectory from a broken one; the boundary of the one-dimensional moduli space is the set of once-broken trajectories, giving $\partial^2 = 0$. *Sources:* Floer, CMP 118 (1988), JDG 28 (1988), CMP 120 (1989); Hofer–Salamon (1995); Salamon's Park City lectures (1999); Audin–Damian's book (2014), the most complete exposition and the one to formalize. *Needs:* S1-type analysis on cylinders, E1, E2 (Fredholm theory of the linearized operator, spectral flow), transversality (Floer–Hofer–Salamon 1995). *Needed by:* Floer homology, the Arnold conjecture (in the monotone / semipositive case). **L–XL.**

S7. Taubes: the Weinstein conjecture in dimension three. Every contact form on a closed 3-manifold has a periodic Reeb orbit. *Proof:* solutions of the Seiberg–Witten equations with a large parameter r exist by Kronheimer–Mrowka's nonvanishing of monopole Floer homology, and as $r \to \infty$ the spinor's zero set concentrates on a union of Reeb orbits. *Source:* Taubes, Geom. Topol. 11 (2007). *Needs:* SW, S3-type asymptotic analysis, monopole Floer homology (Kronheimer– Mrowka's book, XL). **XL.**

F. Foundations of moduli spaces

F1. sc-calculus. An sc-Banach space is a Banach space $E = E_0$ with a decreasing sequence of Banach subspaces $E_0 \supseteq E_1 \supseteq \cdots$, each inclusion $E_{m+1} \to E_m$ compact and $E_\infty = \bigcap E_m$ dense in each E_m. A map $f : U \to F$ between open subsets of sc-Banach spaces is sc^0 if it maps $U_m := U \cap E_m$ continuously to F_m for all m; sc^1 if for each $x \in U_1$ there is a bounded linear $Df(x) : E_0 \to F_0$ with $\|f(x+h) - f(x) - Df(x)h\|_{F_0} = o(\|h\|_{E_1})$, and the tangent map $Tf : TU \to TF$, $Tf(x,h) = (f(x), Df(x)h)$ on $TU = U_1 \oplus E_0$ (the shifted scale) is sc^0; sc^∞ by iteration. Theorems: the sc-chain rule; sc-smooth maps that are classically smooth level-wise; the basic examples (shift maps, action of reparametrization on $W^{k,p}$ scales). *Sources:* HWZ, JEMS 9 (2007), §2; HWZ, *Polyfold and Fredholm Theory* (2021), Ch. 1; Fabert–Fish–Golovko–Wehrheim, *Polyfolds: a first and second look* (EMS Surv. 2016), the friendliest entry point. *Needs:* Banach spaces, compact operators (in Mathlib). *Needed by:* F2–F4. *Notes:* the ideal opening target of Track II: no geometry, definitions are the content, the chain rule's proof is genuinely delicate (it uses the compactness of the inclusions and the shift of levels), and the classical counterexamples (the reparametrization action is not C^1 on a single Banach space but is sc-smooth on a scale) can be formalized as sanity checks. **S–M.**

F2. sc-retractions, splicings, M-polyfolds. An sc-smooth retraction $r : U \to U$ ($r \circ r = r$) has image $O = r(U)$, a *local model*; the tangent space $T_x O =$

im $Dr(x)$ can change dimension from point to point; sc-smoothness of maps between local models is well defined (independent of the retraction); M-polyfolds are spaces locally modelled on such O, with boundary and corners; splicings as the standard way to build retractions from a family of projections (the gluing profile). *Sources:* HWZ (2007) §3–4; HWZ, *sc-smoothness, retractions and new models for smooth spaces*, DCDS 28 (2010); the 2021 book Ch. 2. *Needs:* F1. *Needed by:* F3, F4. *Note:* the independence of the smooth structure from the retraction is the key theorem and the natural second milestone. **M–L.**

F3. Polyfold Fredholm theory. Strong bundles over M-polyfolds; sc-Fredholm sections (defined via a filled-in local normal form: after a change of coordinates, the section is a classical Fredholm map plus a "contraction germ"); the sc-implicit function theorem (a regular sc-Fredholm section has a solution set that is a finite-dimensional manifold with boundary and corners, of the expected dimension); properness and compactness of solution sets; the perturbation theorem (for a proper sc-Fredholm section with compact zero set there exist sc^+-multisection perturbations making the section transverse, with the perturbed zero set a compact weighted branched orbifold, unique up to cobordism); orientations. *Sources:* HWZ, GAFA 19 (2009) (implicit function theorems); Geom. Topol. 13 (2009) (Fredholm functors, polyfolds); *Integration theory on the zero sets of polyfold Fredholm sections*, Math. Ann. 346 (2010); the 2021 book Parts II–III. *Needs:* F1, F2, E2, Sard–Smale, orbifold-type quotients (ep-groupoids). *Needed by:* F4 and its applications: the Arnold conjecture in full generality via polyfolds (Filippenko–Wehrheim 2022), Gromov–Witten invariants for general symplectic manifolds. *Notes:* this is the body of theory whose independent verification the field would most welcome; it is long (the book is about 1,000 pages), uniform in style, and its difficulties are of bookkeeping and estimates rather than of ideas, which is the profile a machine handles best. **L–XL.**

F4. Polyfolds of Gromov–Witten theory and of SFT. The space of stable maps of a given genus and homology class into a compact symplectic manifold carries the structure of an M-polyfold (more precisely, an ep-groupoid) built from splicings that implement gluing at nodes with an exponential gluing profile; the Cauchy–Riemann operator is an sc-Fredholm section of a strong bundle over it; its zero set is the compact Gromov–Witten moduli space; F3 then gives the virtual fundamental class and the Gromov–Witten invariants. For SFT, the same with holomorphic buildings. *Sources:* HWZ, *Applications of polyfold theory I: the polyfolds of Gromov–Witten theory*, Mem. AMS 248 (2017); Fish–Hofer, *Lectures on polyfolds and symplectic field theory* (2018) and the sequence of papers towards the SFT polyfolds. *Needs:* F3, S1 (resp. S4), Deligne–Mumford. **XL.**

F5. The alternative frameworks. *Kuranishi structures* (Fukaya–Ono, Topology 38, 1999; FOOO, *Lagrangian intersection Floer theory*, AMS/IP 2009; FOOO, *Kuranishi structures and virtual fundamental chains*, Springer 2020): finite-dimensional charts (V, E, Γ, s, ψ) with an obstruction bundle, coordinate changes, and a virtual fundamental chain built from multisections. *Kuranishi atlases* (McDuff–Wehrheim, Geom. Topol. 21 (2017), two papers): a reformula-

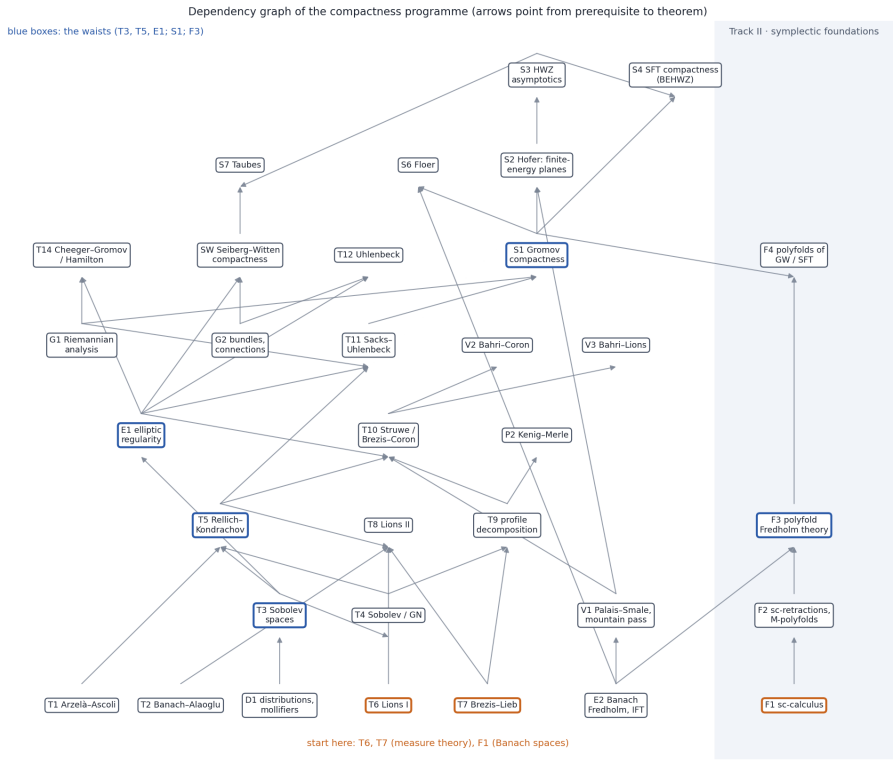
tion designed to expose and repair the topological difficulties in the original approach. *Implicit atlases* (Pardon, Geom. Topol. 20 (2016); Pardon, Geom. Topol. 23 (2019) for contact homology): a homological formulation in which the virtual fundamental class is constructed sheaf-theoretically from local finite-dimensional reductions, avoiding the need for a global smooth structure. Also Li–Tian, Liu– Tian, Ruan, Siebert (1996–99) for the semipositive-free virtual cycle, and Ionel–Parker's relative theory. *Needs:* S1 and gluing analysis; algebraic topology (Čech cohomology, Steenrod homology). *Notes:* the history of these frameworks (1996–2020), including the public debate over the completeness of the early proofs and the decade of re-exposition that followed, is the strongest argument in this document for machine verification: the community settled the question by consensus, and a checked version of any one of these frameworks would replace consensus with certainty. Pardon's is the most algebraic and may be the easiest to formalize once S1 and gluing are in place; polyfolds are the one whose first half (F1–F3) can begin today. **L–XL each.**

G. A count

Fifty-odd entries: 3 already in Mathlib (T1, T2, T2′), 6 of size S or S–M (D1, E2, T6, T7, F1, and Ekeland's principle inside V1), 13 of size M or M–L, 16 of size L or L–XL, and the rest XL. The S-sized entries are exactly the two beginnings recommended in §4: the measure-theoretic concentration-compactness lemma and the sc-calculus.

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3. The dependency graph



The graph shows the main entries of the tally (not every primed variant). It has two waists. In the analytic track, **T3 (Sobolev spaces) and T5 (Rellich–Kondrachov)** are on the path to everything above them, and **E1 (elliptic regularity)** is on the path to everything geometric. In the symplectic track, **S1 (Gromov compactness)** is on the path to everything in blocks E and F except the sc-calculus itself, while **F1–F2 (sc-calculus and M-polyfolds)** depend on nothing but Banach space theory and can be started on day one.

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4. Order of attack: two tracks

Track I: analysis

Milestone 0 (weeks): the measure-theoretic core. Formalize T6 and T7. Both are short, both are used everywhere, and both can be stated and proved with what Mathlib has. Lions's lemma I is a theorem about sequences of probability measures on $\mathbb{R}^n$ with the concentration function $Q_k(R) = \sup_y \mu_k(B(y,R))$; its proof is a diagonal argument and a splitting by a cutoff, exactly what a

proof assistant handles well and humans get wrong in the details. Success is a first machine-checked concentration-compactness statement, and a calibration of cost.

Milestone 1 (months): Sobolev spaces on $\mathbb{R}^n$ and bounded domains. T3, T4, T5, and the Banach-space Fredholm and implicit function theory E2 (much of which is a matter of bundling what Mathlib has). Design decisions to make once and early:

- Functions versus equivalence classes: Mathlib's L^p spaces are quotients; define weak derivatives on the quotient via the distributional pairing and $W^{k,p}$ as a subtype of L^p.
- Domains: begin with $\mathbb{R}^n$ and open sets; prove extension for Lipschitz domains later.
- Rellich–Kondrachov via the Kolmogorov–Riesz–Fréchet criterion for compactness in L^p, a reusable theorem in its own right.

Milestone 2 (months): variational compactness. V1 (mountain pass with the deformation lemma), T8, then T9. These need only Milestone 1 and are the theorems most often called "well known" and least often written down in full.

Milestone 3 (a year): elliptic regularity and the first bubbling theorems. E1 for the semilinear equation $-\Delta u = |u|^{2^*-2}u$, then T10 (Struwe; Brezis–Coron). This is the first theorem a geometer recognizes as geometric analysis. **Then V2, Bahri–Coron.** It is the natural first target in the critical-points-at-infinity line because it is a complete, celebrated theorem whose proof is an explicit computation of the topology of the set of bubble configurations, and the gluing and expansion estimates it needs are exactly the kind of long, error-prone analysis where a machine check has value. V3 (Bahri–Lions) follows with the same tools plus spectral theory.

Milestone 4 (years): manifolds and gauge theory. G1, then T11 (Sacks–Uhlenbeck), then G2 and **SW before T12**: Seiberg–Witten compactness is a maximum-principle argument and is the cheapest compactness theorem in gauge theory, while Uhlenbeck's needs the full Coulomb gauge machinery. T14 and GMT are separate branches.

Track II: symplectic and contact topology

Milestone S0 (weeks to months): the sc-calculus. F1 needs nothing but Banach spaces: sc-Banach spaces are decreasing scales $E_0 \supseteq E_1 \supseteq \cdots$ with compact inclusions; sc-continuity, sc^1 and sc-smoothness are definitions; the sc-chain rule is the first theorem, and it is a theorem with a subtle proof (the tangent map lives on the shifted scale). This is a perfect first formalization target: it is the foundation of a large theory, its definitions are the content, it is short, and it has never been checked by machine. It also provides a calibration of how well agents handle Hofer–Wysocki–Zehnder's style of argument before committing to F3.

Milestone S1 (months): sc-retractions and M-polyfolds. F2. The key results are that sc-smooth retractions have well-defined tangent spaces (which vary in dimension), that the models are stable under sc-diffeomorphism, and the basic splicing constructions. Still no geometry: this is functional analysis with a new smoothness notion.

Milestone S2 (a year or more): polyfold Fredholm theory. F3: strong bundles, sc-Fredholm sections, the sc-implicit function theorem, and the perturbation theorem. This is the heart of the 2021 book. Formalizing it would settle, by a means everyone accepts, a body of foundations that the field relies on and that only a handful of people have read in full. It is also where the machine's advantages are largest: the arguments are long, uniform, and combinatorially intricate (boundary and corner strata, the "gluing profile"), and not conceptually deep in the way S1 is.

Milestone S3 (years): the geometric input. S1 (Gromov compactness) first, then F4 for Gromov–Witten theory (HWZ 2017), which is the first place where compactness (S1) and the Fredholm theory (F3) meet and produce an invariant. Then S2 and S3 (Hofer's finite-energy planes and the HWZ asymptotics), which are the entry to the contact world, and S4 (SFT compactness) as the summit of the compactness track. S6 (Floer) can run in parallel once S1-type analysis on cylinders exists.

On the alternative frameworks (F5). Kuranishi structures and implicit atlases are as legitimate as polyfolds, and a lab whose interest is Lagrangian Floer theory rather than SFT should weigh Fukaya–Oh–Ohta–Ono's or Pardon's route. Pardon's implicit atlases are the most algebraic and may formalize most cleanly once S1 and gluing are in hand; polyfolds have the advantage that their first half (F1–F3) needs no geometry at all and can begin today. In any case the analytic core (S1 and gluing) is shared, and it is the expensive part.

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5. How to run it

Statements first, checked by a human, then proofs by agents. The lesson of the Milnor project that accompanies this document: the theorem statements were the bottleneck and the proofs were not. The human mathematician's job is to write, in ordinary language, exactly what each theorem says, including the hypotheses that textbooks omit (which subsequence, which topology, which normalization, which gluing profile); the formal statement is then written and read back in prose for sign-off; only then are proofs generated. A Lean blueprint (a dependency graph of statements with informal proofs) is the right artefact, and it is what a lab without compute can produce and a lab with compute cannot produce without a mathematician.

State over abstract spaces, not $\mathbb{R}^n$ with coordinates. Sard's theorem in the Milnor project was stated for arbitrary finite-dimensional normed spaces and Haar

measures because the induction passes to hyperplanes. Sobolev theory and the sc-calculus present the same choice; the sc-calculus in particular is *defined* over abstract Banach scales and should stay there.

Choose the proof that formalizes, not the proof in the textbook. Milnor's cube counting for Sard's Step 3 became "Hölder maps divide Hausdorff dimension", which Mathlib had. Expect the same: Kolmogorov–Riesz instead of diagonal extraction; Brezis–Lieb from Fatou and uniform integrability; profile decompositions by induction on the energy defect; for Gromov compactness, Hummel's or McDuff–Salamon's exposition rather than the original.

Expect the work to be in the infrastructure. Two thirds of the Milnor project's lines were infrastructure Mathlib lacked. For Track I it is Sobolev spaces and elliptic regularity; for Track II it is Riemann surfaces with their degenerations and the analysis of the Cauchy–Riemann operator on punctured surfaces. Plan, budget, and publish these as library contributions in their own right.

What is not worth doing first. Cheeger–Gromov, SFT compactness, Uhlenbeck compactness and the polyfolds of SFT are the most famous entries and the worst places to start: each needs a large body of geometry that does not yet exist in any library. A group that begins at the top will spend its first years on preliminaries. A group that begins at the bottom (T6, T7, F1) will have checked theorems in weeks, a checked bubbling theorem and a checked sc-calculus within a year, and will have built, along the way, what everyone else needs.

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6. What the mathematician can do without compute

1. Write the blueprint for Milestones 0–1 and S0–S1: precise statements of T3–T7 and F1–F2 with all hypotheses, informal proofs chosen for formalizability, and the dependency graph.
2. Decide the design questions of §4 (quotients versus functions; domains; the compactness criterion; the definition of sc-smoothness to adopt from the several equivalent ones in the literature) and record the reasons.
3. Write the prose statements a formalizer will be asked to match, in the format used in the chapters of this project: a theorem, a list of what is silently assumed, and what it is for.
4. Pick the headline theorem each track should deliver first with a physical or geometric meaning: for Track I, Bahri–Coron or the existence of a ground state for a nonlinear Schrödinger equation; for Track II, the polyfold Fredholm theory itself, whose verification the field would welcome.

That is a document of perhaps eighty pages. It is the part of the project that requires a mathematician and not a machine, and it is what the labs do not have.

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7. Selected references

Bahri, *Critical points at infinity in some variational problems* (1989). Bahri–Coron, CPAM 41 (1988). Bahri–Lions, CPAM 45 (1992). Bahri–Rabinowitz, Ann. IHP 8 (1991). Bahri, *Flow lines and algebraic invariants in contact form geometry* (2003). Lions, Ann. IHP 1 (1984), Rev. Mat. Iberoam. 1 (1985). Struwe, Math. Z. 187 (1984). Brezis–Coron, CPAM 37 (1984) and CMP 92 (1985). Sacks– Uhlenbeck, Ann. Math. 113 (1981). Uhlenbeck, CMP 83 (1982), two papers. Gérard, ESAIM COCV 3 (1998). Bahouri–Gérard, Amer. J. Math. 121 (1999). Kenig–Merle, Invent. Math. 166 (2006). Gromov, Invent. Math. 82 (1985). Hofer, Invent. Math. 114 (1993). Hofer–Wysocki–Zehnder, Ann. IHP 13 (1996), GAFA 5 (1995), and sequels. Bourgeois–Eliashberg–Hofer–Wysocki–Zehnder, Geom. Topol. 7 (2003). Hofer–Wysocki–Zehnder, JEMS 9 (2007), GAFA 19 (2009), Geom. Topol. 13 (2009); *Polyfold and Fredholm Theory*, Springer (2021); *Applications of polyfold theory I*, Mem. AMS (2017). Fish–Hofer, *Lectures on polyfolds and symplectic field theory* (2018). Fukaya–Oh–Ohta–Ono, *Lagrangian intersection Floer theory* (2009), *Kuranishi structures and virtual fundamental chains* (2020). McDuff–Wehrheim, Geom. Topol. 21 (2017). Pardon, Geom. Topol. 20 (2016). Taubes, Geom. Topol. 11 (2007). Hofer– Zehnder, *Symplectic invariants and Hamiltonian dynamics* (1994). Federer–Fleming, Ann. Math. 72 (1960). Allard, Ann. Math. 95 (1972). Hamilton, Amer. J. Math. 117 (1995).